

Intermediate Macro

Chapter 1, 2: Introduction

Instructor: Mihyun You

Department of Economics
Georgetown University

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ECON102: Intermediate Macro

- Instructor: Mihyun You
E-mail: my555@georgetown.edu
Office Hours: By appointment (Zoom)
- Class Hours: MTWR 1:10PM-3:05PM
- Class Room: ICC 105
- Main textbook: Blanchard, Oliver, *Macroeconomics*, 8th Edition, Pearson, 2021.
- Prerequisites: ECON001 and ECON002 (or ECON003), MATH035

Grading Policy

Final grade will be calculated as a weighted average of problem sets, exams, and class participation:

- **55%** of the grade: $\max\{Exam1, Exam2\}$.
- **35%** of the grade: $\min\{Exam1, Exam2\}$.
- **10%** of the grade: two problem sets (5% each).
- Class participation (curve-up)

Problem Sets and Exams

- Problem Sets

Two problem sets throughout the semester will be graded based on completeness. Working together is encouraged.

Due date: each exam day.

- Exams

Exam 1: July 26, 2022 (Tuesday)

Exam 2: August 11, 2022 (Thursday)

Course Policies

- Eating and drinking are NOT allowed during class.
- Wearing masks follows the university policy.

Road Map to Macroeconomics

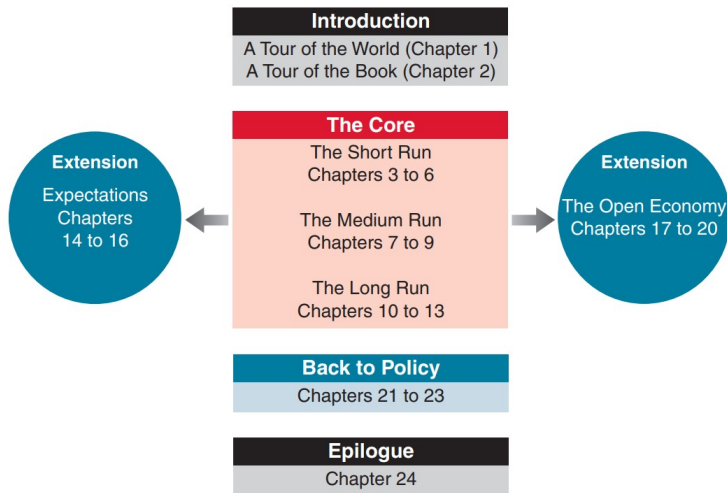


Table of Contents

- ① Course Description
- ② Chapter 1: A Tour of the World
- ③ Chapter 2: A Tour of the Book
 - 2.1 Aggregate Output
 - 2.2 The Unemployment Rate
 - 2.3 The Inflation Rate
 - 2.4 Okun's Law and the Phillips Curve
 - 2.5 The Short, the Medium, and the Long Run



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What is Macroeconomics?

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- The best way to answer this question is to take you on an economic tour of the world.

Issues of Macroeconomics

- The big issues triggered by the crisis: What caused the crisis? Why was it transmitted from the United States to the rest of the world? In retrospect, what could and should have been done to prevent it? Were the monetary and fiscal responses appropriate? Why has the recovery been so slow in Europe? How was China able to maintain high growth during the crisis?

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- Study and learn from real everyday life.
- News articles are good resources of study material.
 - ① WEO (World Economic Outlook):
<https://www.imf.org/en/Publications/WEO>
 - ② OECD Economic Outlook:
<https://www.oecd.org/economic-outlook/>
 - ③ The Economist: <https://www.economist.com/>

America's inflation headache gets worse



Finance & economics | Gaining altitude

America's inflation headache gets worse

As prices soar, markets tumble—and the Democrats may be next



Jun 13th 2022 (Updated Jun 14th 2022) | WASHINGTON, DC

Takeaway

"Economies, like people, get sick – high unemployment, recessions, financial crises, low growth. Macroeconomics is about **why it happens**, and **what can be done** about it."

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Key Tools and Concepts

- Chapter 2 introduces **index numbers**.
- This chapter defines formally the basic macroeconomic concepts of **nominal** and **real gross domestic product** (GDP), **GDP growth**, the **GDP deflator**, the **unemployment rate**, the **consumer price index** (CPI), and the **inflation rate**, as well as associated concepts such as **value added**, **intermediate inputs**, the **labor force**, and the **participation rate**.
- This chapter distinguishes the **short run**, the **medium run**, and the **long run** time frame. The distinction establishes the basic theoretical framework.

Three Central Macroeconomic Variables

- The three central macroeconomic variables are **output**, **unemployment**, and **inflation**.
- ① How do economists define output, the unemployment rate, and the inflation rate, and why do economists care about these variables?
 - ② How do these three variables relate to each other?
 - ③ What factors affect output in the short run, the medium run, and long run?

GDP: Production and Income

2.1 Aggregate Output

- The measure of **aggregate output** in the national income accounts is called the **gross domestic product**, or **GDP**.
- GDP has three equivalent definitions:
 - ① the value of final goods and services produced during a given period
 - ② the sum of value added during a given period
 - ③ the sum of labor and capital income and indirect taxes.

GDP example

2.1 Aggregate Output

Steel Company (Firm 1)

Revenues from sales	\$100
Expenses	\$80
Wages	\$80
Profit	\$20

Car Company (Firm 2)

Revenues from sales	\$200
Expenses	\$170
Wages	\$70
Steel purchases	\$100
Profit	\$30

Nominal and Real GDP

2.1 Aggregate Output

- **Nominal GDP**, $\$Y_t$ is the sum of the quantities of final goods produced times their *current price*.
- **Real GDP**, Y_t is the sum of the quantities of final goods times *constant prices*.

Year	Quantity of Cars	Price of Cars	Nominal GDP	Real GDP (in 2012 dollars)
2011	10	\$20,000	\$200,000	\$240,000
2012	12	\$24,000	\$288,000	\$288,000
2013	13	\$26,000	\$338,000	\$312,000

Practical Issues regarding Real GDP

2.1 Aggregate Output

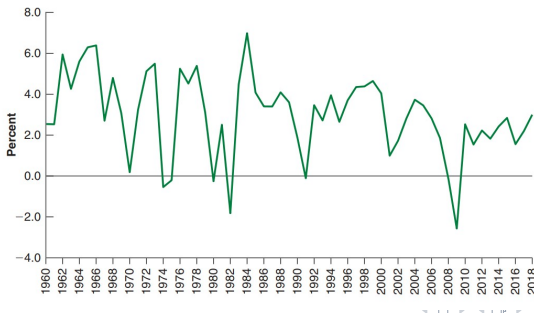
- The construction of real GDP involves two complications:
 - ① The economy produces many goods. One must decide how to weight the value of the output of each good to produce aggregate real GDP.
→ Chain-weighting technique allows the relative price of goods to change over time.
 - ② The quality of similar goods changes over time.
→ Hedonic pricing estimates the market value of a good's characteristics—speed, durability, and so on,

GDP: Level versus Growth Rate

2.1 Aggregate Output

- In assessing the performance of the economy from year to year, economists focus on the rate of growth of real GDP, often called just **GDP growth**.

$$\text{GDP growth in year } t = \frac{Y_t - Y_{t-1}}{Y_{t-1}} \approx \log Y_t - \log Y_{t-1}.$$



Construction of the Unemployment Rate

2.2 The Unemployment Rate

- The working age population is defined as those aged 15 to 64.
- $$\underset{\text{labor force}}{L} = \underset{\text{employment}}{N} + \underset{\text{unemployment}}{U}.$$
- participation rate = $\frac{\text{labor force } (L)}{\text{working age population}}.$
- unemployment rate, $u = \frac{U}{L}.$
- An unemployed person must satisfy two conditions:
 - ① he or she does not have a job, and
 - ② he or she is looking for one.
- Note that those who do not have a job but are not looking for one are counted as **not in the labor force**. Those who give up looking for a job are *discouraged workers*.

Labor Shortage Problem in U.S. Economy

2.2 The Unemployment Rate

- Unemployment rates are falling in the U.S., but labor shortages persist. An article in "The Atlantic" points out that about 7 million workers have left their jobs despite rising wages and a surging U.S. economy.
(<https://www.youtube.com/watch?v=eC92E4FkAcs>)



The GDP Deflator

2.3 The Inflation Rate

- The inflation rate is the growth rate of the aggregate price level.
- GDP deflator, $P_t = \frac{\text{Nominal } GDP_t}{\text{Real } GDP_t} = \frac{\$Y_t}{Y_t}$.
- Since the choice of base year is arbitrary, the level of the GDP deflator is meaningless. The rate of change of the GDP deflator, however, is meaningful; it is one measure of inflation.
- $\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}$.
- $\$Y_t = P_t Y_t$.
→ The growth rate of nominal GDP is equal to the inflation rate plus the growth rate of real GDP.

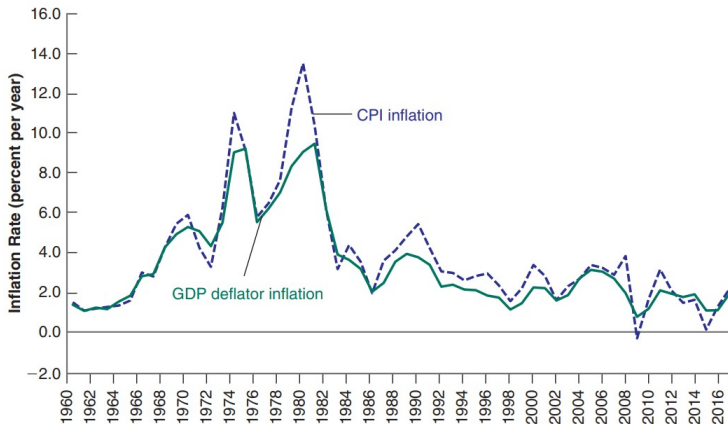
The Consumer Price Index

2.3 The Inflation Rate

- The GDP deflator gives the average price of output—the final goods *produced* in the economy. But consumers care about the average price of consumption—the goods they *consume*.
- An alternative measure of the price level is the Consumer Price Index (CPI)—another index number.
- CPI represents the cost of living. Each month, Bureau of Labor Statistics (BLS) employees visit stores to find out what has happened to the price of the goods on the specific list.

GDP Deflator and CPI

2.3 The Inflation Rate



Why Do Economists Care about Inflation?

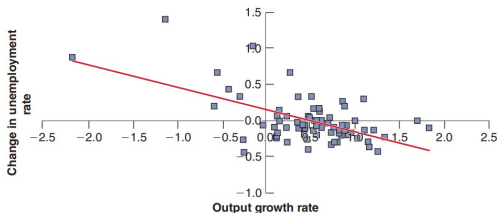
2.3 The Inflation Rate

- During periods of inflation, not all prices and wages rise proportionately. Inflation affects income distribution.
- Variations in relative prices also lead to more uncertainty, making it harder for economic agents to make decisions about the future.
- What about deflation? If inflation is so bad, does this imply that deflation (negative inflation) is good?

Okun's Law

2.4. Okun's Law and the Phillips Curve

- Intuition suggests that if output growth is high, unemployment will decrease.

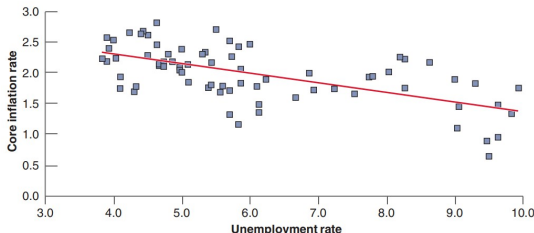


- The slope of the regression line is -0.3. This line crosses the horizontal axis at the point where the quarterly output growth is roughly equal to 0.5%, equivalently when annual output growth is equal to 2%.

The Phillips Curve

2.4 Okun's Law and the Phillips Curve

- When unemployment becomes very low, the economy is likely to overheat, and that this will lead to upward pressure on inflation.



- The regression line is also downward sloping, although the fit is not as good as it was for Okun's law.

Three Time Frames in Macroeconomics

2.5 The Short, Medium, and the Long Run

- What determines the level of aggregate output in an economy?
- In the **short run**, year-to-year movements in output are primarily driven by movements in **demand**. To focus on the role of demand, we assume that firms are willing to supply any quantity at a given price (No supply constraints).
- In the **medium run**, the economy tends to return to the level of output determined by the **supply factors**: the capital stock, the level of technology, and the size of the labor force. Over a the **decade** or so, these factors move sufficiently slowly that we can take them as given.
- In the the **long run**, neither technology, nor capital, nor skills are given. Over a **few decades or more**, the true determinants of output are factors like a country's education system, its saving rate, and the quality of its government.